A Fixed Confidence Gate for Selective Classification

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Abstract

We examine a fixed confidence gate for selective classification using a synthetic evaluation fixture. The gate abstains from predictions below a specified confidence threshold. We report accuracy among accepted predictions alongside coverage on two synthetic datasets. Conditional accuracy increases on one dataset and decreases on the other; these observations do not establish an overall accuracy improvement. A range bound for selective risk requires positive coverage and bounded loss. Statistical uncertainty, runtime, and distribution shift remain unevaluated.

1. Introduction

Selective classification allows a model to abstain from uncertain predictions. We study a fixed confidence gate, rather than a learned rejection policy. This synthetic evaluation asks whether a fixed threshold changes accuracy among accepted predictions. Coverage is reported alongside conditional accuracy because the gate changes which predictions are evaluated.

2. Method

Gate uses a fixed classifier and accepts a prediction when its maximum class probability is at least 0.8; otherwise, it abstains. Always uses the same classifier and accepts every prediction. Neither method retrains the model. For a fixed classifier h and acceptance function $a(x) \in \{0, 1\}$, assume $0 \leq \ell(h(x), y) \leq 1$ and positive coverage $C = \mathbb{E}[a(X)] > 0$. Selective risk is

$$R_{\text{sel}} = \frac{\mathbb{E}[a(X)\ell(h(X), Y)]}{C}. \tag{1}$$

Theorem 1 *Under these assumptions,* $0 \leq R_{\text{sel}} \leq 1$.

The result is a range bound, not a generalization guarantee or a comparison between Gate and Always. Appendix A gives the full proof.

3. Synthetic Evaluation

Alpha and Beta each contain 1000 synthetic samples. Both methods use the same model and 5 evaluation seeds; results are seed means, with no variance estimates. Table 1 shows a higher conditional accuracy on Alpha and a lower conditional accuracy on Beta. Different evaluation populations prevent interpreting these differences as an overall accuracy improvement. No significance tests, runtime measurements, or distribution-shift evaluations were performed.

Table 1. Synthetic fixture values. Always accuracy uses all samples; Gate accuracy is conditional on acceptance. These values are not research findings.

Dataset	Always (%)	Gate (%)	Coverage (%)
Alpha	80.0	84.0	60.0
Beta	76.0	75.5	58.0

4. Limitations and Conclusion

A single baseline and synthetic datasets do not establish state-of-the-art performance. The classifier and data-generation details are missing and required for reproducibility. The fixed threshold does not guarantee lower risk, and deployment benefits remain untested. Formatting uses the official author guidelines (for, 2026).

References

Icml 2026 author instructions. https://icml.cc/Conferences/2026/AuthorInstructions, 2026.

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A. Proof of the Range Bound

Since $0 \leq a(X)\ell(h(X), Y) \leq a(X)$, taking expectations gives $0 \leq \mathbb{E}[a(X)\ell(h(X), Y)] \leq C$. Division by $C > 0$ proves Theorem 1. This argument applies to the specified distribution; it does not establish a bound under distribution shift.